Akshat Kankani

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Overview

Computer Science undergraduate specializing in **Artificial Intelligence**, with strong interests in **systems programming, NLP, and low-level software design**. Experienced in building **end-to-end machine learning pipelines** and **Unix-style system software from scratch**, with a focus on correctness, modularity, and real-world behavior. Actively pursuing depth-oriented projects spanning NLP systems, operating systems, and developer tooling.

Education

Manipal Institute of Technology — Bengaluru, KA
Bachelor of Technology in Computer Science (Artificial Intelligence) — *July 2023 – May 2027*

- CGPA: 8.01

Projects

BiasScope | *Python, PyTorch, HuggingFace, Streamlit*

- Built an end-to-end NLP platform to analyze public opinion by scraping online forums and news sources based on user-defined prompts
- Designed and implemented a multi-stage NLP pipeline including text normalization, transformer-based summarization, sentiment analysis, and bias detection
- Implemented ensemble model pipelines combining multiple pretrained and fine-tuned models to improve robustness and accuracy
- Computed sentiment divergence and bias scores by comparing public discourse, media narratives, and AI-generated perspectives
- Developed a low-latency REST API for model inference and data retrieval
- Built a responsive Streamlit frontend with real-time visualization and seamless API integration
- Designed a PostgreSQL-backed persistence layer for storing scraped data, processed outputs, and user queries

Custom UNIX-like File System | *C*

- Implemented a block-based file system from scratch using inode-based metadata management
- Designed direct, single-indirect, and double-indirect block addressing for scalable file storage
- Implemented hierarchical directory structures with full path resolution support
- Supported core file operations including create, read, write, append, and delete
- Built mount and unmount mechanisms using a disk image abstraction with persistent metadata
- Focused on on-disk consistency, correctness of metadata updates, and modular design for extensibility

Unix Shell Implementation | *C++*

- Built a Unix-style shell using low-level POSIX system calls
- Implemented robust command parsing and tokenization with argument handling
- Developed built-in commands including `cd`, `pwd`, and `exit`
- Added support for pipelines with correct file descriptor chaining
- Added support for TAB auto-completion, command history and history persistence
- Implemented I/O redirection (`>`, `>>`, `<`) with file creation and append behavior
- Supported background process execution and process lifecycle management using `fork`, `exec`, and `wait`
- Handled signals and child processes to ensure shell stability and correct job control

Technical Skills

Languages: C, C++, Python
Databases: mySQL, PostgreSQL
Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks, Google Colab
Libraries: NumPy, pandas, Matplotlib, scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers, Streamlit